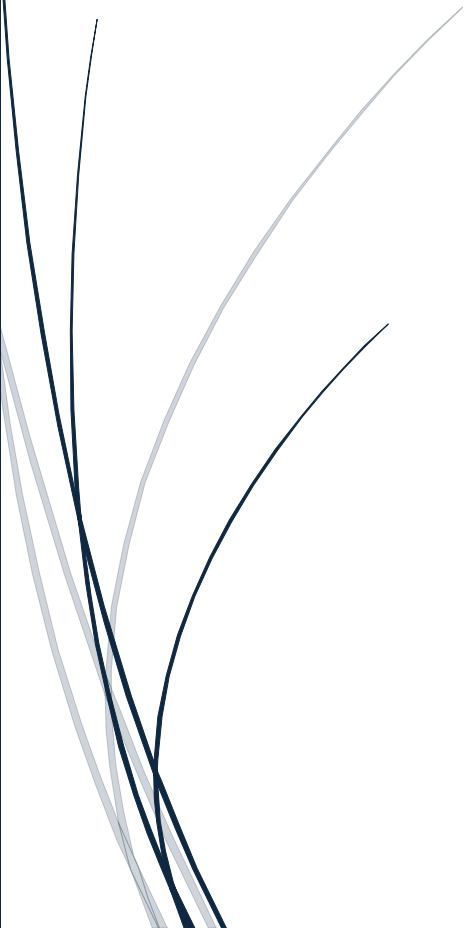


7/26/2026

[ShadowFox Data Science Internship]

[Advanced Level Task]

[Netflix Movies & TV Shows Data Analysis
Using Python]



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Introduction:

Netflix is one of the world's largest and most popular online streaming platforms, offering a wide variety of movies, TV shows, documentaries, and original content across different countries and languages. Since its launch, Netflix has transformed the entertainment industry by providing users with on-demand digital content that can be accessed anytime and anywhere.

In this project, Python is used as the primary programming language along with powerful data analysis and visualization libraries such as **Pandas**, **NumPy**, **Matplotlib**, and **Seaborn**. The dataset is cleaned and preprocessed before performing Exploratory Data Analysis (EDA). Various charts and graphs are created to visualize patterns, compare different categories, and identify meaningful insights.

Objectives:

The primary objectives of this project are:

1. To analyze the Netflix Movies and TV Shows dataset using Python.
2. To perform data cleaning and preprocessing for accurate analysis.
3. To compare the number of Movies and TV Shows available on Netflix.
4. To identify the countries contributing the highest amount of Netflix content.
5. To analyze the trend of content released over the years.
6. To study the distribution of content ratings and genres.

Scope of the Project:

This project focuses on analyzing the Netflix Movies and TV Shows dataset using Python. The analysis includes data cleaning, handling missing values, exploratory data analysis (EDA), and data visualization. Different charts and graphs are used to understand content distribution, release trends, country-wise production, ratings, genres, and other important characteristics of the dataset.

The project is limited to the available dataset and does not include real-time Netflix data. However, the analysis demonstrates practical data science techniques that can be applied to similar real-world datasets. The findings can help understand streaming content patterns and improve decision-making based on historical data.

Research Questions:

The project aims to answer the following research questions:

1. How many Movies and TV Shows are available on Netflix?
2. Which country has produced the highest number of Netflix titles?
3. Which year has the highest number of content releases?
4. Which content ratings are most common?

And etc....

Dataset Description:

The dataset used in this project is **Netflix Movies and TV Shows Dataset**. It contains information about movies and TV shows available on Netflix.

The dataset includes the following important attributes:

- **show_id** – Unique identifier for each title.
- **type** – Specifies whether the content is a Movie or TV Show.
- **title** – Name of the movie or TV show.
- **director** – Director of the content.
- **cast** – Main actors and actresses.
- **country** – Country where the content was produced.
- **date_added** – Date when the content was added to Netflix.
- **release_year** – Year of release.
- **rating** – Audience rating (TV-MA, PG, TV-14, etc.).
- **duration** – Duration of movies or number of seasons for TV shows.
- **listed_in** – Genre or category of the content.
- **description** – Short summary of the content.

Tools and Technologies Used:

The following tools and technologies were used to complete this project:

- Python
- Google Colab
- Pandas
- NumPy
- Matplotlib
- Seaborn
- MS Word
- GitHub

Import Libraries, Data Loading, Cleaning and EDA:

Netflix Movies and TV Shows Data Analysis Using Python

ShadowFox Data Science Internship

Advanced Level Task

Submitted By: Priyanshu Gond

Import Libraries:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
plt.style.use('ggplot')
```

Uploaded CSV File:

```
from google.colab import files
uploaded = files.upload()
```

Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

Saving netflix_titles.csv to netflix_titles.csv

First Five Rows of the Dataset:

```
import pandas as pd
df =
pd.read_csv("netflix_titles.csv")
df.head()
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
0	1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	PG-13	90 min	Documentaries	As her father nears the end

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
												of his life, filmm..
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thabana...	South Africa	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town...
2	s3	TV Show	Ganglands	Julien Leclecq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021	TV-MA	1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train I...

Last Five Rows of the Dataset:

```
df.tail()
```


	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
8802	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	November 20, 2019	2007	R	158 min	Cult Movies, Dramas, Thrillers	A political cartoonist, a crime reporter and a...
8803	s8804	TV Show	Zombie Dumb	NaN	NaN	NaN	July 1, 2019	2018	TV-Y7	2 Seasons	Kids' TV, Korean TV Shows, TV Comedies	While living alone in a spooky town, a young g...
8804	s8805	Movie	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody Harrelson, Emma Stone, ...	United States	November 1, 2019	2009	R	88 min	Comedies, Horror Movies	Looking to survive in a world taken over by zo...
8805	s8806	Movie	Zoom	Peter Hewitt	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States	January 11, 2020	2006	PG	88 min	Children & Family Movies, Comedies	Dragged from civilian life, a former superhero...
8806	s8807	Movie	Zubaan	Moze Singh	Vicky Kaushal, Sarah-Jane Dias, Raaghav	India	March 2, 2019	2015	TV-14	111 min	Dramas, International Movies, Music &	A scrappy but poor boy worms his way

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
					Chana n...						Musical s	into a ty...

Dataset Information:

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   show_id               8807 non-null   object
1   type                  8807 non-null   object
2   title                 8807 non-null   object
3   director              6173 non-null   object
4   cast                  7982 non-null   object
5   country               7976 non-null   object
6   date_added            8797 non-null   object
7   release_year          8807 non-null   int64
8   rating                8803 non-null   object
9   duration              8804 non-null   object
10  listed_in             8807 non-null   object
11  description            8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
```

Dataset Shape:

```
df.shape
```

```
(8807, 12)
```

Dataset Columns:

```
df.columns
```

```
Index(['show_id', 'type', 'title',  
      'director', 'cast', 'country',  
      'date_added',  
            'release_year', 'rating',  
      'duration', 'listed_in', 'description'],  
      dtype='object')
```

Missing Values:

```
df.isnull().sum()
```

	0
show_id	0
type	0
title	0
director	2634
cast	825
country	831
date_added	10
release_year	0
rating	4
duration	3
listed_in	0
description	0

dtype: int64

Statistical Summary:

```
df.describe(include="all")
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
count	8807	8807	8807	6173	7982	7976	8797	8807.000000	8803	8804	8807	8807
unique	8807	2	8807	4528	7692	748	1767	NaN	17	220	514	8775
top	s8807	Movie	Zubaaan	Rajiv Chilaka	David Attenborough	United States	January 1, 2020	NaN	TV-MA	1 Season	Dramas, International Movies	Paranormal activity at a lush, abandoned prope...
freq	1	6131	1	19	19	2818	109	NaN	3207	1793	362	4
mean	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2014.180198	NaN	NaN	NaN	NaN
std	NaN	NaN	NaN	NaN	NaN	NaN	NaN	8.819312	NaN	NaN	NaN	NaN
min	NaN	NaN	NaN	NaN	NaN	NaN	NaN	1925.000000	NaN	NaN	NaN	NaN
25%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2013.000000	NaN	NaN	NaN	NaN
50%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2017.000000	NaN	NaN	NaN	NaN
75%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2019.000000	NaN	NaN	NaN	NaN
max	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2021.000000	NaN	NaN	NaN	NaN

Check duplicate rows:

```
df.duplicated().sum()
```

```
np.int64(0)
```

Remove duplicate rows:

```
df.drop_duplicates(inplace=True)
```

Fill missing values:

```
df["director"] =  
df["director"].fillna("Unknown")  
df["cast"] = df["cast"].fillna("Unknown")  
df["country"] =  
df["country"].fillna("Unknown")  
df["rating"] = df["rating"].fillna("Not  
Rated")
```

Data Visualization (Graph) :

Movies vs TV Shows (Count Plot) Code

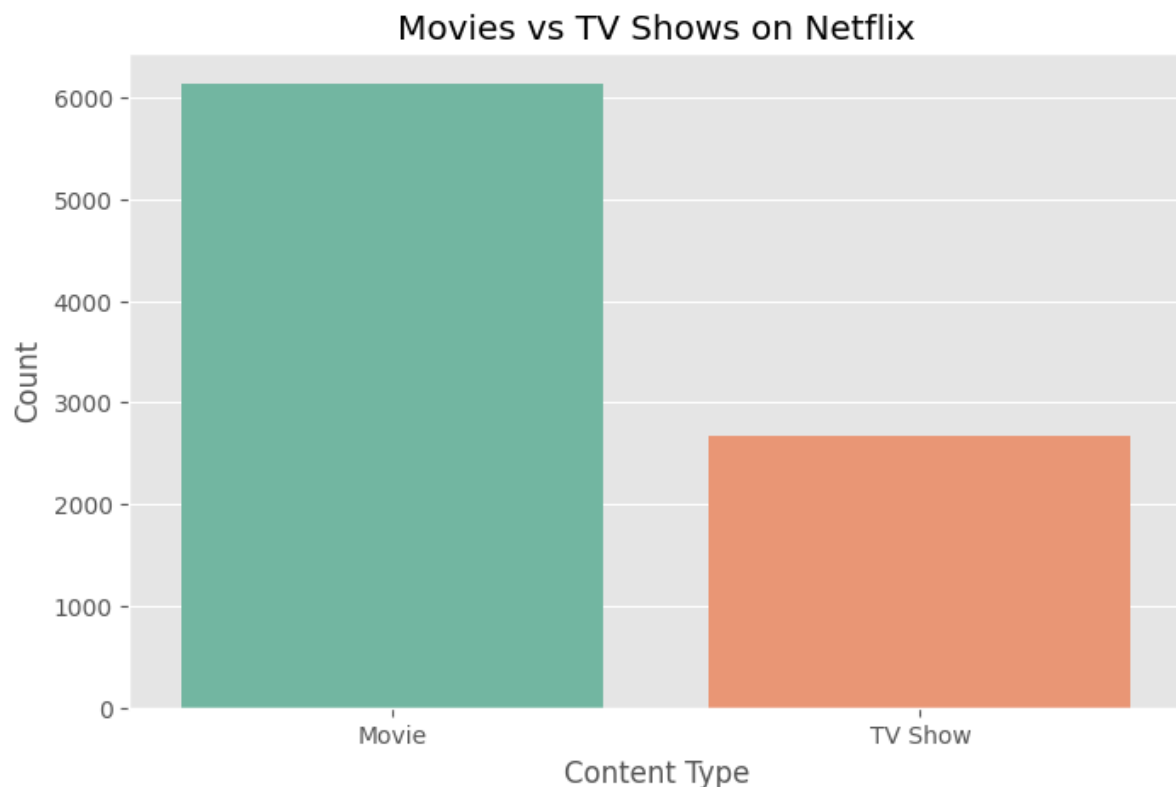
```
plt.figure(figsize=(8,5))  
sns.countplot(x='type', data=df,  
palette='Set2')
```

```
plt.title("Movies vs TV Shows on  
Netflix")  
plt.xlabel("Content Type")  
plt.ylabel("Count")  
plt.show()
```

/tmp/ipykernel_1626/1490833205.py:4:
FutureWarning:

Passing `palette` without assigning `hue`
is deprecated and will be removed in
v0.14.0. Assign the `x` variable to `hue`
and set `legend=False` for the same
effect.

```
sns.countplot(x='type', data=df,  
palette='Set2')
```



Explanation:

This graph compares the total number of Movies and TV Shows available on Netflix.

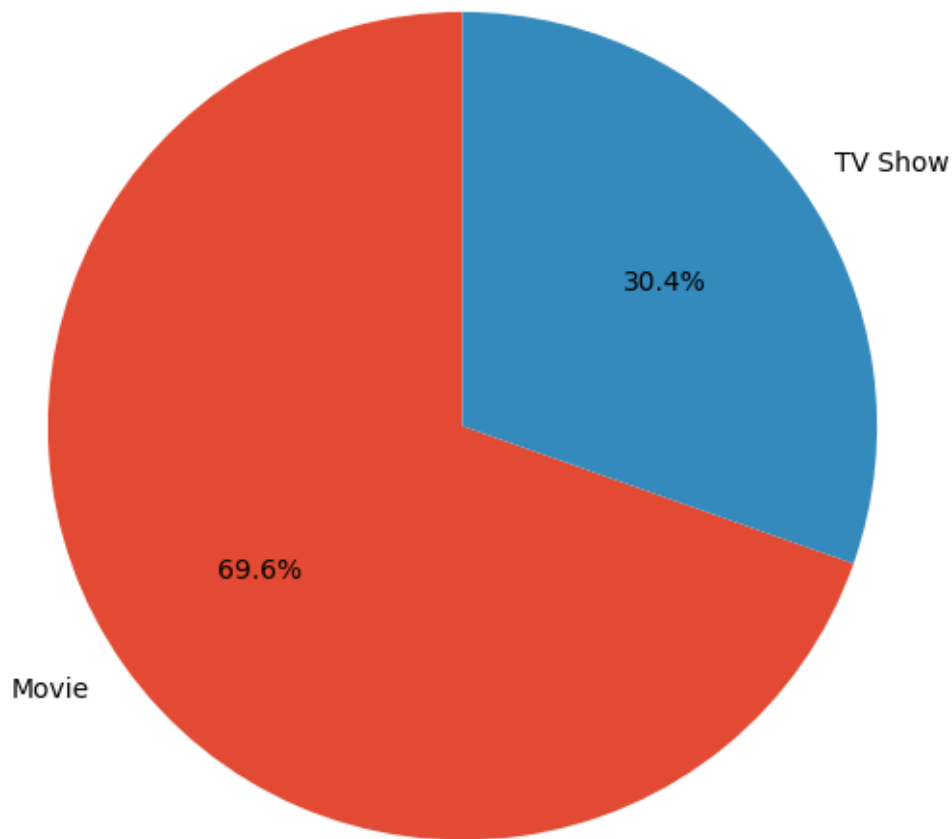
The graph clearly shows which content type dominates Netflix's library.

If Movies have a higher count than TV Shows, mention that Netflix focuses more on movie content.

Movies vs TV Shows (Pie Chart) Code

```
plt.figure(figsize=(7,7))
df['type'].value_counts().plot(
    kind='pie',
    autopct='%1.1f%%',
    startangle=90
)
plt.title("Percentage of Movies and TV Shows")
plt.ylabel("")
plt.show()
```

Percentage of Movies and TV Shows



Explanation:

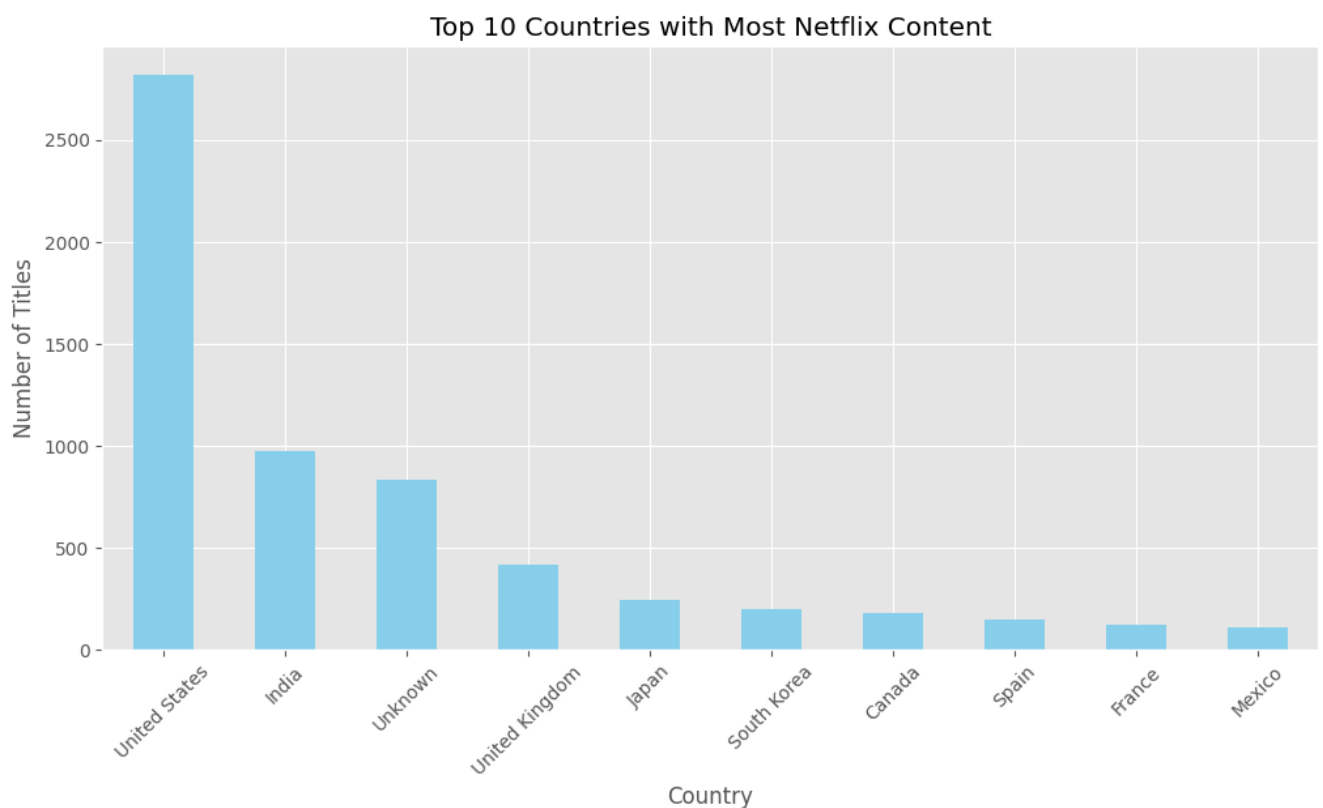
This pie chart represents the percentage distribution of Movies and TV Shows available on Netflix.

Top 10 Countries Code

```
plt.figure(figsize=(12, 6))
df['country'].value_counts().head(10).plot(
    kind='bar',
    color='skyblue'
```



```
)  
plt.title("Top 10 Countries with Most  
Netflix Content")  
plt.xlabel("Country")  
plt.ylabel("Number of Titles")  
plt.xticks(rotation=45)  
plt.show()
```

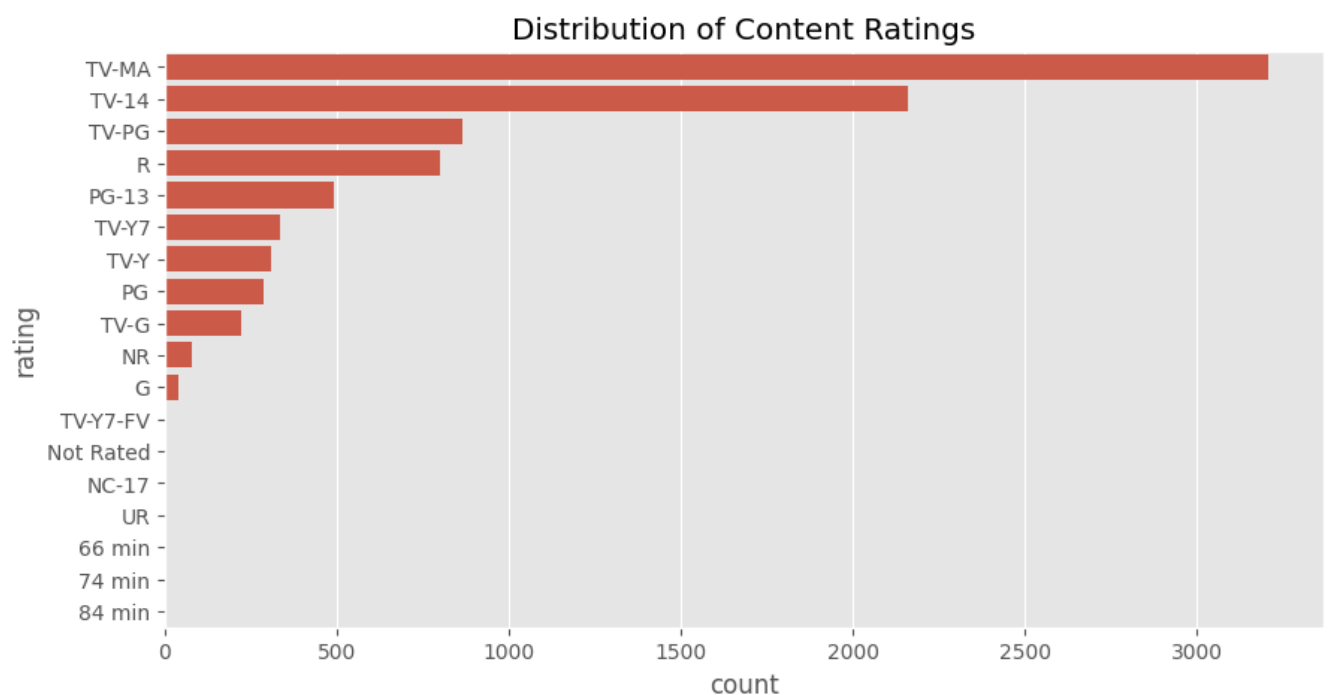


Explanation:

This graph identifies the countries contributing the highest number of Movies and TV Shows to Netflix.

Content Rating Distribution Code

```
plt.figure(figsize=(10,5))
sns.countplot(
    y='rating',
    data=df,
    order=df['rating'].value_counts().ind
ex
)
plt.title("Distribution of Content
Ratings")
plt.show()
```

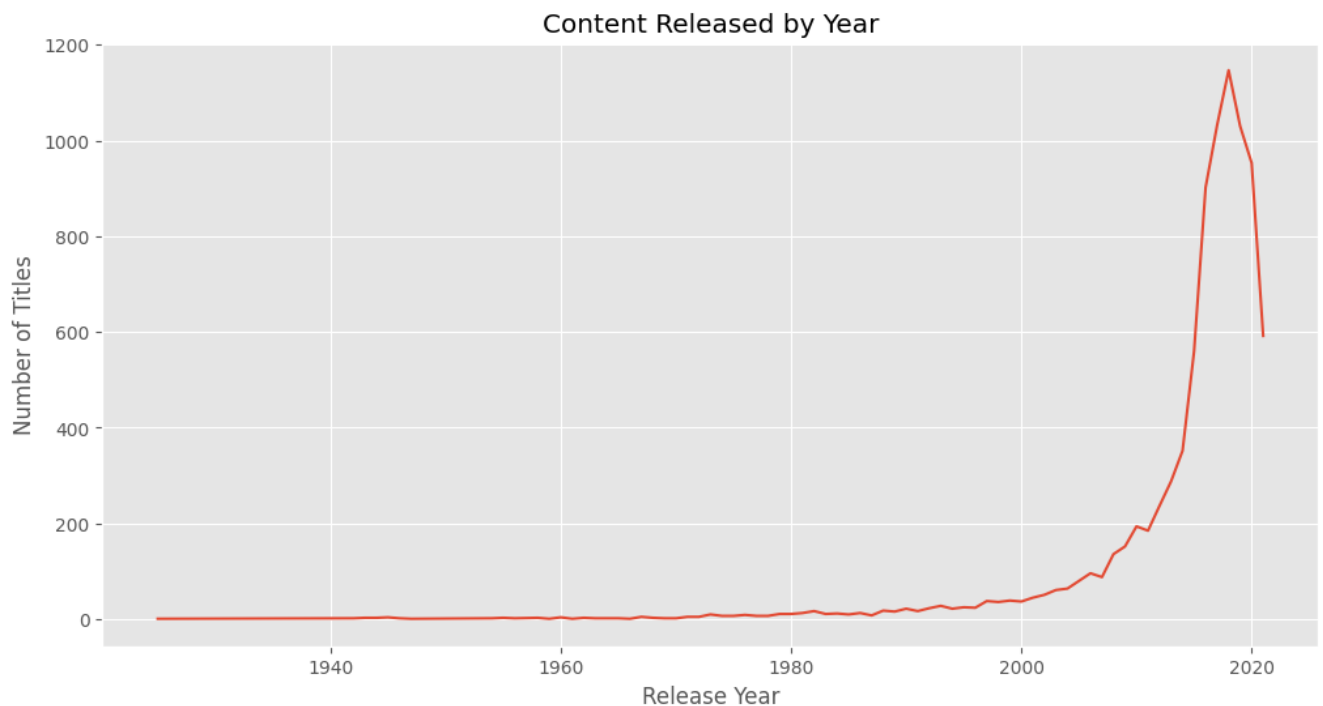


Explanation:

This graph displays the distribution of different content ratings available on Netflix, such as **TV-MA**, **TV-14**, **PG**, **PG-13**, **R**, etc.

Content Released by Year Code

```
plt.figure(figsize=(12,6))
df['release_year'].value_counts().sort_index().plot()
plt.title("Content Released by Year")
plt.xlabel("Release Year")
plt.ylabel("Number of Titles")
plt.show()
```

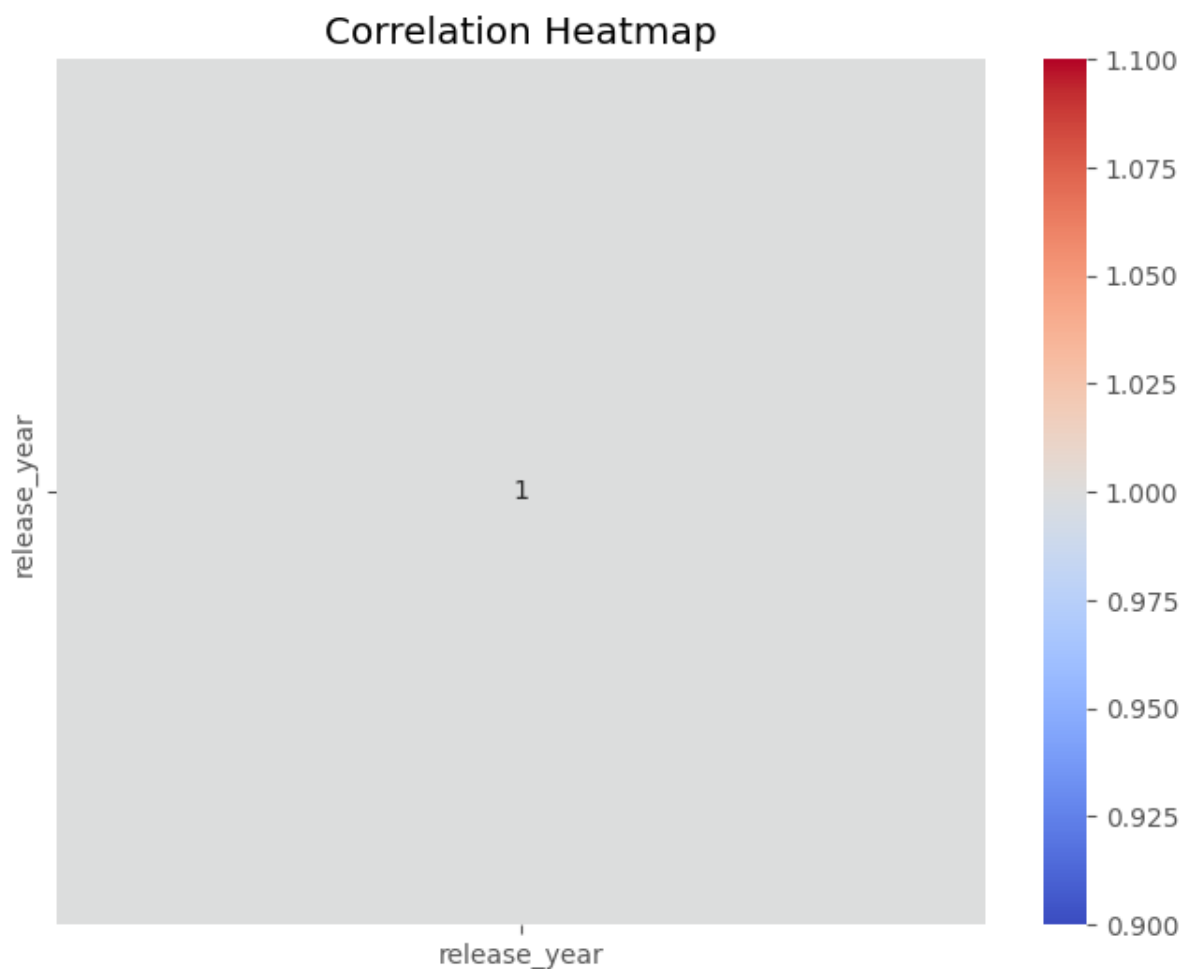


Explanation:

This graph is shown by the Content Released by Year to the Netflix Movies & TV shows.

Heatmap Code

```
numeric_df =  
df.select_dtypes(include=['int64', 'float64'])  
plt.figure(figsize=(8, 6))  
sns.heatmap(  
    numeric_df.corr(),  
    annot=True,  
    cmap='coolwarm'  
)  
plt.title("Correlation Heatmap")  
plt.show()
```



Key Insights:

- Netflix contains more Movies than TV Shows.
- The United States contributes the highest number of titles.
- Netflix content has increased significantly in recent years.
- TV-MA and TV-14 are among the most common content ratings.
- Drama and International Movies are popular genres.
- Some directors have contributed multiple titles to the platform.
- Movie durations are mostly concentrated within a common runtime range.
- Data visualization helps identify trends and patterns efficiently.

Discussion:

The analysis of the Netflix Movies and TV Shows dataset provides valuable insights into the content available on the platform. The exploratory data analysis (EDA) reveals that Netflix offers a larger number of Movies compared to TV Shows, indicating that movies form a significant portion of its content library.

Country-wise analysis shows that a few countries contribute a major share of Netflix content, with the United States being one of the leading contributors. The release year analysis highlights a significant increase in content production over recent years, reflecting Netflix's rapid expansion and growing investment in original programming.

Conclusion:

This project successfully analyzed the Netflix Movies and TV Shows dataset using Python. The dataset was cleaned, processed, and explored using various data analysis techniques. Multiple visualizations were created to identify trends related to content type, release year, country, ratings, genres, directors, and movie duration.

The analysis showed that Netflix contains a higher number of Movies than TV Shows and that the platform has experienced significant growth over the years. The visualizations also highlighted the most active countries, common content ratings, and popular genres.

Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn proved to be effective tools for performing data analysis and creating informative visualizations. This project demonstrates the importance of data visualization in understanding large datasets and supporting data-driven decision-making.

Future Scope:

The project can be extended in several ways:

- Develop machine learning models to recommend movies and TV shows.
- Perform sentiment analysis using user reviews.
- Build an interactive dashboard using Streamlit or Power BI.
- Analyze trends across different streaming platforms such as Amazon Prime Video and Disney+.

References:

- Netflix Movies and TV Shows Dataset (Kaggle)
- Python Official Documentation
- Pandas Documentation
- NumPy Documentation
- Matplotlib Documentation
- Seaborn Documentation
- Google Colab Documentation
- GitHub Documentation